REPORT 3

Sales Prediction Report

Machine learning models for daily store sales forecasting

Project	Rossmann Store Sales Prediction
Task	Task 3 - Sales Prediction
Selected model	Tuned Random Forest Regressor
Report date	12 November 2026

1,177.08	794.05	0.8516	41,088
RMSE (best model)	MAE (best model)	R^2 score (best model)	Test predictions generated

1. Project Overview

Objective: To predict daily store sales using Machine Learning models based on historical sales and store information.

2. Dataset Information

Item	Value
Training records	1,017,209
Testing records	41,088
Store records	1,115
Training features	27
Training date range	2013-01-01 to 2015-07-31

3. Data Preprocessing

- Cleaned datasets were used.
- Training and testing datasets were merged with store data.
- Missing values were checked.
- Categorical variables were encoded.
- Numerical features were retained.

4. Feature Engineering

- Year
- Month
- Day
- Day of Year
- Week of Year
- Weekend Indicator
- Month Start and Month End
- Competition Age
- Promotion Activity

5. Model Development

The following Machine Learning models were developed:

- Linear Regression
- Random Forest Regressor
- Tuned Random Forest Regressor

6. Model Evaluation

Evaluation metrics: Root Mean Squared Error (RMSE), Mean Absolute Error (MAE) and R^2 Score.

Model	RMSE	MAE	R^2
Tuned Random Forest	1,177.08	794.05	0.85
Random Forest	1,363.03	928.15	0.80
Linear Regression	2,664.04	1,946.24	0.24

7. Best Model

Selected Model	RMSE	MAE	R^2 Score
Tuned Random Forest	1,177.08	794.05	0.8516

8. Hyperparameter Tuning

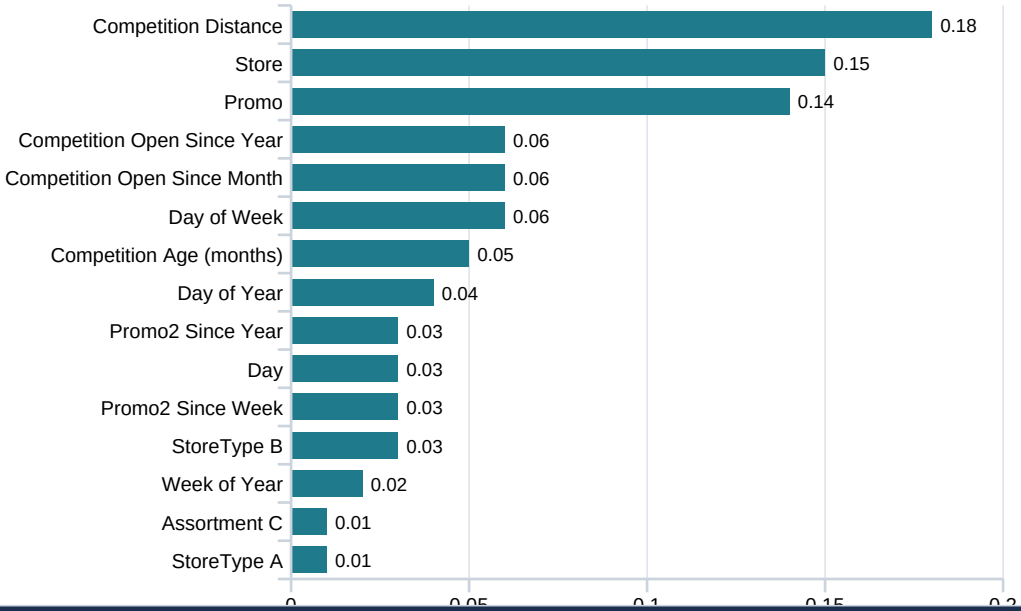
RandomizedSearchCV was used for Random Forest hyperparameter optimisation.

Best Parameter	Value
n_estimators	80
min_samples_split	5
min_samples_leaf	2
max_features	0.8
max_depth	25

Best cross-validation RMSE: 2,175.26

9. Feature Importance

Top 15 features ranked by importance:



Rank	Feature	Type	Importance
1	Competition Distance	Numerical	0.18
2	Store	Numerical	0.15
3	Promo	Numerical	0.14
4	Competition Open Since Year	Numerical	0.06
5	Competition Open Since Month	Numerical	0.06
6	Day of Week	Numerical	0.06
7	Competition Age (months)	Numerical	0.05
8	Day of Year	Numerical	0.04
9	Promo2 Since Year	Numerical	0.03
10	Day	Numerical	0.03
11	Promo2 Since Week	Numerical	0.03
12	StoreType B	Categorical	0.03
13	Week of Year	Numerical	0.02
14	Assortment C	Categorical	0.01
15	StoreType A	Categorical	0.01

10. Validation Predictions

Total validation predictions: 175,084. Actual and predicted sales were compared to evaluate model performance.

11. Final Test Predictions

Total test predictions: 41,088. Final sales predictions were generated for the test dataset.

12. Output Files

Models	`linear_regression_model.pkl`
	`random_forest_model.pkl`
	`tuned_random_forest_model.pkl`
	`best_sales_prediction_model.pkl`

Predictions	`model_comparison.csv`
	`validation_predictions.csv`
	`final_sales_predictions.csv`
	`feature_importance.csv`

Visualisations	`01_model_comparison.png`
	`02_actual_vs_predicted.png`
	`03_residual_distribution.png`
	`04_feature_importance.png`

Reports	`Sales_Prediction_Report.txt` `Sales_Prediction_Report.pdf`

13. Conclusion

Three Machine Learning models were developed and evaluated for Rossmann store sales prediction. The best-performing model was selected based on validation performance. Feature importance analysis was performed to identify important sales prediction factors, and final predictions were generated for the test dataset.